

Spillover Effects of Payouts on Asset Prices and Real Investment

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Background: Demand Shocks and Asset Prices

- ▶ If demand curves for stocks slope downward, then investor-specific demand shocks drive prices and can be used as instruments for...
 1. **Corporate Finance:** Exog. variation in discount rates, Tobin's Q etc.
 2. **Asset Pricing:** Market efficiency tests, arbitrage, realized versus expected returns, instruments in demand systems etc.
 3. **Macro Finance:** Impact of asset purchases etc.
- ▶ Standard requirement: Demand should be *exogenous* ($\perp$ to fundamentals) and *relevant* (large enough to affect prices).
 1. **Cross-section:** Index inclusions (Shleifer '86), fire sales (Coval & Stafford '07), flows (Lou '12), mandates (Kojien & Yogo '19)
 2. **Market level:** GIV (Gabaix & Kojien '22), dividends (Hartzmark & Salomon '22)

This Paper

1. Construct new demand shock for the cross-section:
Funds reinvest dividends in co-held stocks
2. Test impact at daily frequency:
1% demand shock raises prices by 0.8% (permanently?)
3. Aggregate to annual frequency:
1% demand shock raises prices by 2bp
4. Corporate finance: (on the last 2 pages)
Use demand shock as instrument in investment and capital structure regressions

My Comments

1. Construct new demand shock for the cross-section: **great** ✓
 - ▶ Relevant and exogenous!
 - ▶ Available for all stocks at all times (unlike index inclusions/fire sales)
 - ▶ Arguably better than ordinary flow-induced trading
2. Test impact at daily frequency: ✓ + a few minor comments
 - ▶ Persistence
 - ▶ Magnitude of the price impact
 - ▶ Expected versus unexpected

3. Aggregate to annual frequency:
How do demand shocks at higher frequency add up over time? From micro-structure to macro asset pricing
4. Corporate finance:
Maybe in a separate paper?

Advertisement: Why is this instrument great?

- ▶ Ordinary Flow Induced Trading (FIT):

$$\text{FIT}_t(n) = \sum_i \text{Flow}_{i,t} w_{i,t-1}(n)$$

..uses flow from households to mutual funds

Drawbacks: Endogenous if flows are not pure noise (reflect HH preferences about underlying) or chase returns at higher frequency

- ▶ Payout Induced Trading (PIT)

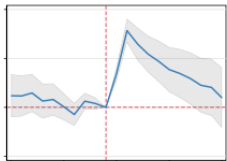
$$\text{PIT}_t(n) = \sum_i \text{Payout-Flow}_{i,t} w_{i,t-1}(n)$$

...uses flow from payouts of *other* stocks in *i*'s portfolio.

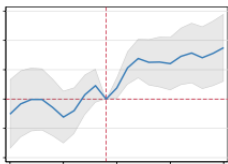
Benefits: Plausibly exogenous, available at high frequency

Comment 1. Persistence

1. Impact from own dividend reinvestment reverts



2. Impact from co-held stock's dividend reinvestment does not revert



Why are they so different? If market makers read your paper, would you expect reversal? Or is reinvestment in co-held stocks slower/risky?

Comment 2. Magnitude of the price impact

$$\text{Return}_t = \alpha + \beta \text{Demand Shock}_t + \epsilon_t$$

Interpreting the OLS regression coefficient *structurally* depends critically on measurement of demand shock. Estimated price multiplier is $\beta = 0.8$ (demand-elasticity $\frac{1}{\beta} = 1.25$), assuming...

1. ... 100% reinvestment of dividend in co-held stocks.
Empirically, reinvestment in co-held stocks is 60% (Table 2). This suggests a price multiplier of $\frac{0.8}{0.6} = 1.33$ (demand-elasticity of 0.75)
2. ... demand is scaled by 13F shares (not by shares outstanding).
13F institutions hold on average 60% of shares outstanding (great variation in this number). If the other 40% do not reinvest in co-held stocks this suggests a price multiplier of $0.8 \times 0.6 = 0.48$ (and a demand-elasticity of 2.1).

Comment 3. Expected versus unexpected demand shocks

- ▶ If investors are forward looking, expected demand shocks should drive prices today (Gabaix and Koijen, 2022)

$$\text{Return}_t = E_t \left[\sum_{s=t}^{\infty} \beta(s, t) \text{Demand Shock}_s \right] + \epsilon_t$$

- ▶ The date and amount of dividend payments (\$1 Trillion /Year) are perfectly predictable. Why do market participants require such large price concessions (\$0.8 Trillion /Year)?
- ▶ What's the sharpe ratio on a strategy that front-runs the reinvestment impact?
- ▶ Maybe reinvestment in co-held stocks is not very predictable? E.g. happens slowly, subject to household choice etc.
- ▶ Suggestion: Use daily holdings data from ETFs to show when reinvestment happens and how predictable it is.

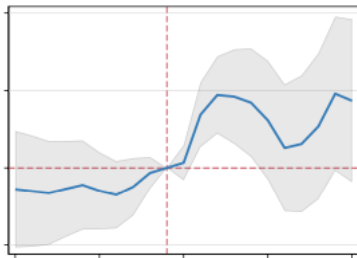
Conclusion

Great paper!

- ▶ This seems like a nice alternative to Russel reconstitutions (Chang, Hong & Liskovitch (2015)) with far more *treated* stocks.
- ▶ Why extend to corporate finance? Demand shock is cool, very useful to many streams of literature *by itself*

Other comments (1)

- ▶ Placebo Test: Impact for stock dividends (as opposed to cash dividends)



- ▶ Significant impact. Almost as large as impact from cash dividend. Is this evidence against the proposed mechanism of reinvested cash? Why not?
- ▶ Do your Investment-Q results change when using FIT instead of PIT?

Other comments (2)

- ▶ Impact is much lower at annual frequency
 - ▶ Daily: $\frac{\% \text{Daily Return}}{\% \text{Daily Shock}} = 0.8$
 - ▶ Annual: $\frac{\% \text{Annual Return}}{\% \text{Annual Shock}} = 0.02$
- ▶ Potential Explanation: You aggregate demand shocks over a year, but prices partly revert. Seems like an interesting setting to explore how high frequency demand shocks propagate through time.
- ▶ More generally: Annual impact is far way from prior estimates at quarterly frequency Lou (2012), Pavlova & Sikorskaya, 2022, Chang, Hong & Liskovitch (2015)
- ▶ Also: How can you detect 2 basis points at an annual stock vol of 20%?